| Sample numbers in normal order | | | Sample numbers after bit reversal | |
|---|---|---|---|---|
| *Decimal* | *Binary* | | *Decimal* | *Binary* |
| 0 | 0000 | | 0 | 0000 |
| 1 | 0001 | | 8 | 1000 |
| 2 | 0010 | | 4 | 0100 |
| 3 | 0011 | | 12 | 1100 |
| 4 | 0100 | | 2 | 0010 |
| 5 | 0101 | | 10 | 1010 |
| 6 | 0110 | ⇨ | 6 | 0100 |
| 7 | 0111 | | 14 | 1110 |
| 8 | 1000 | | 1 | 0001 |
| 9 | 1001 | | 9 | 1001 |
| 10 | 1010 | | 5 | 0101 |
| 11 | 1011 | | 13 | 1101 |
| 12 | 1100 | | 3 | 0011 |
| 13 | 1101 | | 11 | 1011 |
| 14 | 1110 | | 7 | 0111 |
| 15 | 1111 | | 15 | 1111 |

FIGURE 12-3
The FFT bit reversal sorting. The FFT time domain decomposition can be implemented by sorting the samples according to bit reversed order.

separate stages. The first stage breaks the 16 point signal into two signals each consisting of 8 points. The second stage decomposes the data into four signals of 4 points. This pattern continues until there are $N$ signals composed of a single point. An **interlaced decomposition** is used each time a signal is broken in two, that is, the signal is separated into its even and odd numbered samples. The best way to understand this is by inspecting Fig. 12-2 until you grasp the pattern. There are $Log_2 N$ stages required in this decomposition, i.e., a 16 point signal ($2^4$) requires 4 stages, a 512 point signal ($2^7$) requires 7 stages, a 4096 point signal ($2^{12}$) requires 12 stages, etc. Remember this value, $Log_2 N$; it will be referenced many times in this chapter.

Now that you understand the structure of the decomposition, it can be greatly simplified. The decomposition is nothing more than a *reordering* of the samples in the signal. Figure 12-3 shows the rearrangement pattern required. On the left, the sample numbers of the original signal are listed along with their binary equivalents. On the right, the rearranged sample numbers are listed, also along with their binary equivalents. The important idea is that the binary numbers are the *reversals* of each other. For example, sample 3 (0011) is exchanged with sample number 12 (1100). Likewise, sample number 14 (1110) is swapped with sample number 7 (0111), and so forth. The FFT time domain decomposition is usually carried out by a **bit reversal sorting** algorithm. This involves rearranging the order of the $N$ time domain samples by counting in binary with the bits flipped left-for-right (such as in the far right column in Fig. 12-3).